

Shared evidence | Greenhouse trial

Greenhouse challenge: grow more with a water budget

Name _____ Date _____

All numerical values are constructed. Same plant species/age, pot size, nutrients and temperature; random group allocation. Dry-mass gain estimated from matched start/end groups of 5 plants after 14 days.

Plan	Light	Water supplied(L/plant)	Mean dry-mass gain(g/plant)
A	Low	0.30	1.8
B	High	0.60	3.0
C	Low	0.15	1.0
D	High	0.15	1.1
E: follow-up	Moderate	0.45	2.6

The brief: mean gain at least 2.5 g per plant, with supplied water at most 0.50 L per plant over 14 days. Supplied water is not exact transpiration. Means do not show individual variation. “Low/moderate/high” are fixed trial settings, not universal growth labels.

W1. Choose greatest gain, then apply budget. W2. Test E against both limits. W3. Request one further check.

Supported route | Check both rules

Use the shared table. Word bank: mean; supplied; budget; photosynthesis; repeat; variation.

S1. Choose the fair light-comparison pair C/D or A/B. Explain what is held equal.

S2. Which A-D plan has greatest mean gain? Why does it fail the full brief?

S3. Test E against both limits using its two values. Calculate its water headroom.

S4. Explain how photosynthesis contributes to biomass. Add one reason a mean cannot guarantee every plant succeeds.

Standard route | Recommend a repeat pilot

Use numerical evidence and a biological explanation.

M1. Choose a fair pair to investigate light. Explain why A/B is not a light-only comparison.

M2. Evaluate A-E against both brief conditions. Recommend a plan or none, with exact evidence.

M3. Explain how light and water relate to photosynthesis and why supplied water is not exact transpiration.

M4. Propose a repeat trial or extra evidence to test the recommendation. State what finding could change your mind.

Stretch route | Test the explanation

Optional Higher-tier interaction challenge for T1. Constructed follow-up uses the same species/conditions: low water 0.15 L, low/high light gains 1.0/1.1 g; higher water 0.60 L, low/high light gains 1.8/3.0 g.

T1. Calculate the light-associated gain difference at each water level. Explain a possible limiting-factor interaction, without claiming the data prove the only mechanism.

T2. Each endpoint dry mass minus the matched starting-group mean gives estimated gains 2.2, 2.5, 2.6, 2.8 and 2.9 g for E. Check their mean. Does a mean at least 2.5 g mean every estimate passes?

T3. An editor ranks plans by gain divided by water supplied. Calculate this ratio for B and E. Explain why the ratio alone cannot replace both brief constraints.

T4. Suggest an improved design to test whether E's performance repeats, controlling a specific factor and recording variation.

Exit and reflection | Evidence can revise a plan

State the decision and its scope.

E1. Which A-D pair changes light while holding supplied water equal?

E2. Explain why B fails the full brief using a number.

E3. Why is E a provisional, mean-based recommendation?

L1. What evidence question did your audience ask? Record a revised or defended recommendation.

Next useful step

- A. Choose a fair comparison.
- B. Apply both conditions.
- C. Explain a plant mechanism.
- D. Inspect variation and limits.